

Kitchen Environment

breakdown of the **coordinates and dimensions** for all major items in the kitchen Unified Robot Description Format (URDF) model, organized by component. This includes positions (XYZ), orientations (RPY), and geometric properties (size/scale) for each item.

1. Table Area

- **Link:** table_area
 - **Position (XYZ):** (-2.7, -0.6, 0) (relative to room_link)
 - **Orientation (RPY):** (0, 0, π) (180° rotation around Z-axis)
 - **Mesh:** big_table_1.dae (visual), big_table_1.stl (collision)
 - **Dimensions** (approximate, inferred from mesh):
 - **Width:** ~1.5m (X-axis)
 - **Depth:** ~0.8m (Y-axis)
 - **Height:** ~0.75m (Z-axis)
 - **Center:** (0, 0, 0.375) (local coordinates)

2. Sink Area

Sink Structure

- **Link:** sink_area
 - **Position (XYZ):** (0.29, 1.03, 0.41) (relative to sink_area_footprint)
 - **Mesh:** SinkArea.dae
 - **Collision Box** (for surface):
 - **Size:** (0.575, 1.6, 0.02) (X, Y, Z)
 - **Center:** (0, -0.225, 0) (local coordinates)

Sink Bowl

- **Link:** sink_area_sink
 - **Position (XYZ):** (0.005, 0.47, 0.4414) (relative to sink_area)
 - **Mesh:** Sink.dae
 - **Dimensions:** Inferred from mesh (no explicit size in URDF).

Drawers (Example: Trash Drawer)

- **Link:** sink_area_trash_drawer_main
 - **Joint:** Prismatic (sliding along X-axis)
 - **Position (XYZ):** (0.01, 0.7, -0.015)
 - **Limits:** lower="0" upper="0.48" (48cm travel)
 - **Mesh:** Drawer_60_58.dae
 - **Handle Position:** (0.3325, 0, 0.2375) (relative to drawer).

3. Oven Area

Oven Structure

- **Link:** oven_area_oven_main
 - **Position (XYZ):** (0.0, 0.0, 0.415) (relative to oven_area_area)
 - **Mesh:** OvenMain.dae
 - **Door Joint:** Revolute (Z-axis)
 - **Position:** (0.2737, 0.0, -0.2614)
 - **Limits:** lower="0" upper="0" (initially locked).

Oven Knobs

- **Example Knob** (oven_area_oven_knob_stove_1):
 - **Position (XYZ):** (0, -0.21, 0) (relative to oven_area_oven_panel)
 - **Orientation (RPY):** (0, $-\pi/2$, 0) (90° around Y-axis)
 - **Mesh:** Knob.dae
 - **Rotation Limits:** 0 to π (180°).

4. Kitchen Island

Island Structure

- **Link:** kitchen_island
 - **Position (XYZ):** (0.2975, 1.1292, 0.421) (relative to kitchen_island_footprint)
 - **Mesh:** IslandArea.dae
 - **Surface Collision Box:**
 - **Size:** (0.8, 2.45, 0.02) (X, Y, Z)
 - **Center:** (0, 0, 0.42) (local coordinates).

Stove

- **Link:** kitchen_island_stove
 - **Position (XYZ):** (0.0, 0.7658, 0.4315)
 - **Mesh:** Stove.dae.

Drawers (Example: Left Upper Drawer)

- **Link:** kitchen_island_left_upper_drawer_main
 - **Joint Position (XYZ):** (0.0025, -0.8, 0.114)
 - **Mesh:** Drawer_60_29.dae
 - **Handle Position:** (0.3325, 0, 0.11).

5. Fridge Area

Fridge Structure

- **Link:** iai_fridge_main
 - **Position (XYZ):** (0, 0, 0) (relative to fridge_area)
 - **Mesh:** Fridge.dae
 - **Door Joint:** Revolute (Z-axis)
 - **Position:** (0.29, 0.3, 0.245)
 - **Limits:** 0 to $\pi/2$ (90°).

Fridge Door Handle

- **Link:** iai_fridge_door_handle
 - **Position (XYZ):** (0.06, -0.57, 0.0)
 - **Mesh:** VHandle90.dae.

6. Global Coordinates Summary

Component	Parent Link	Position (XYZ)	Orientation (RPY)	Dimensions (W×D×H)
Table	room_link	(-2.7, -0.6, 0)	(0, 0, π)	~1.5m × 0.8m × 0.75m
Sink Area	room_link	(1.825, 1.32, 0)	(0, 0, π)	Collision: 0.575×1.6×0.02
Oven Area	room_link	(1.805, 2.52, 0)	(0, 0, π)	Mesh-dependent

Kitchen Island	room_link	(-1.365, 0.59, 0)	(0, 0, 0)	Surface: 0.8×2.45×0.02
Fridge Area	room_link	(1.825, -0.74, 0)	(0, 0, π)	Mesh-dependent

Key Notes:

1. **Positions** are relative to the parent link (e.g., sink_area is child of sink_area_footprint).
2. **Dimensions** are either explicit (collision boxes) or inferred from meshes.
3. **Orientation**: RPY = Roll-Pitch-Yaw (in radians). $\pi = 180^\circ$.
4. **Handles/Drawers**: Prismatic joints define sliding ranges (e.g., 0.48m max for drawers).

Apartment Environment

Breakdown of **coordinates, dimensions, and spatial properties** for an apartment environment modeled in Unified Robot description format (URDF). This includes rooms, furniture, and key interactable objects with their positions, sizes, and geometric relationships.

1. Global Structure

World & Rooms

- **Root Link**: world
- **Rooms** (e.g., living_room_link, bedroom_link, kitchen_link) are connected via fixed joints to world.

2. Living Room

Sofa

- **Link**: sofa
 - **Position**: (2.0, 1.5, 0) (relative to living_room_link)
 - **Orientation**: (0, 0, 0)

- **Collision Box:** (1.8, 0.8, 0.6) (width, depth, height)
- **Center:** (0, 0, 0.3) (local coordinates).

Coffee Table

- **Link:** coffee_table
 - **Position:** (2.0, 0.5, 0)
 - **Collision Box:** (1.0, 0.6, 0.4)
 - **Center:** (0, 0, 0.2).

TV Stand

- **Link:** tv_stand
 - **Position:** (3.0, -1.0, 0)
 - **Collision Box:** (1.5, 0.4, 0.5)
 - **TV Mount Position:** (0, 0, 0.6) (relative to tv_stand).

3. Bedroom

Bed

- **Link:** bed
 - **Position:** (-1.5, 2.0, 0)
 - **Collision Box:** (2.0, 1.5, 0.5)
 - **Center:** (0, 0, 0.25).

Wardrobe

- **Link:** wardrobe
 - **Position:** (-3.0, 2.0, 0)
 - **Collision Box:** (1.0, 0.6, 2.0)
 - **Door Joint:** Revolute (Z-axis, limits 0 to 1.57 radians).

Nightstand

- **Link:** nightstand
 - **Position:** (-1.5, 1.0, 0)
 - **Collision Box:** (0.5, 0.5, 0.5).

4. Kitchen

Refrigerator

- **Link:** fridge
 - **Position:** (0.5, -2.0, 0)
 - **Collision Box:** (0.8, 0.7, 1.8)
 - **Door Joint:** Revolute (Z-axis, limits 0 to 1.57).

Countertop

- **Link:** countertop
 - **Position:** (1.0, -1.5, 0)
 - **Collision Box:** (2.0, 0.6, 0.9).

Sink

- **Link:** kitchen_sink
 - **Position:** (1.2, -1.5, 0.9) (on countertop)
 - **Collision Mesh:** sink.dae (size inferred).

5. Bathroom

Sink

- **Link:** bathroom_sink
 - **Position:** (-2.0, -1.0, 0)
 - **Collision Box:** (0.6, 0.4, 0.8).

Toilet

- **Link:** toilet
 - **Position:** (-2.5, -1.5, 0)
 - **Collision Box:** (0.5, 0.7, 0.8).

Shower

- **Link:** shower
 - **Position:** (-1.5, -2.0, 0)
 - **Collision Box:** (1.0, 1.0, 2.0).

6. Hallway

Entryway Table

- **Link:** entry_table
 - **Position:** (0, 0.5, 0)
 - **Collision Box:** (0.8, 0.3, 0.9).

Closet

- **Link:** hall_closet
 - **Position:** (0.5, 0, 0)
 - **Collision Box:** (1.0, 0.5, 2.0)
 - **Door Joint:** Prismatic (sliding along Y-axis).

7. Spatial Summary Table

Object	Parent Link	Position (XYZ)	Orientation (RPY)	Dimensions (W×D×H)
Sofa	living_room_link	(2.0, 1.5, 0)	(0, 0, 0)	1.8 × 0.8 × 0.6
Bed	bedroom_link	(-1.5, 2.0, 0)	(0, 0, 0)	2.0 × 1.5 × 0.5
Fridge	kitchen_link	(0.5, -2.0, 0)	(0, 0, 0)	0.8 × 0.7 × 1.8
Shower	bathroom_link	(-1.5, -2.0, 0)	(0, 0, 0)	1.0 × 1.0 × 2.0
Entry Table	hallway_link	(0, 0.5, 0)	(0, 0, 0)	0.8 × 0.3 × 0.9

Key Notes:

1. **Positions:** All coordinates are relative to the parent link (e.g., sofa is placed at (2.0, 1.5, 0) in living_room_link).
2. **Collision Boxes:** Explicit dimensions are provided where available; meshes are used otherwise.
3. **Joints:** Doors/drawers use revolute/prismatic joints with defined limits (e.g., fridge door rotates 90°).
4. **Scaling:** Adjust dimensions based on mesh scales if needed (e.g., sink.dae may require scaling).

